Name – Arpita Waghmare

PRN-24070521006

SEC-A

```
%{
#include <stdio.h>
int yylex();
int yyerror(char *);
%}

%token FOR ID NUM ASSIGN RELOP INCDEC

%%
expr : FOR '(' ID ASSIGN NUM ';' ID RELOP NUM ';' ID INCDEC ')' '{' '}' {printf("Valid for loop\n");};
%%

int main()
{
    yyparse();
    return 0;
}

int yyerror(char *s)
{
    printf("Invalid for loop\n");
    return 0;
}
```

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SEC-A

```
%{
#include <stdio.h>
#include "y.tab.h"
%}

%%
"for"                       { return FOR; }
[a-zA-Z_][a-zA-Z0-9_]*      { return ID; }
[0-9]+                      { return NUM; }
"++"|"--"                   { return INCDEC; }
"<"|">"|">="|"<="|"=="|"!=" { return RELOP; }
"="                         { return ASSIGN; }
[();{}]                     { return yytext[0]; }
[ \t\n]+                    {}
.                           {}
%%

int yywrap()
{
    return 1;
}
```

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SEC-A

```
ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ yacc -d prac10.y

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ flex prac10.l

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ gcc lex.yy.c y.tab.c -o a.out

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ ./a.out
for(i=0; i<8 ; i++){}
Valid for loop
for(i=0; i<8 i++)
Invalid for loop

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$
```

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SEC-A

```
%{
#include "y.tab.h"
%}

%%
[0-9]+                      { return NUM; }
[a-zA-Z_][a-zA-Z0-9_]*      { return ID; }
">="|"<="|"=="|"!="|">"|"<" { return RELOP; }
"?"                         { return '?'; }
":"                         { return ':'; }
[ \t\n]+                    {}
.                           {}
%%

int yywrap()
{
    return 1;
}
```

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SEC-A

```
%{
#include <stdio.h>
int yylex();
int yyerror(char *);
%}

%token ID RELOP TRUE FALSE

%%
expr : ID RELOP ID '?' TRUE ':' FALSE ';' {printf("Valid ternary syntax\n");};
%%

int main()
{
    yyparse();
    return 0;
}

int yyerror(char *s)
{
    printf("Invalid ternary syntax\n");
    return 0;
}
```

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SEC-A

```
ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ notepad ternary.l

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ notepad ternary.y

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ yacc -d ternary.y

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ flex ternary.l

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ gcc lex.yy.c y.tab.c -o a.out

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ ./a.out
a > b ? true:false
Invalid ternary syntax


ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/Practical10
$ ./a.out
a > b ? true:false;
Valid ternary syntax
```